Name _____ _____ Baggett/Fall2026/207/Quiz2

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (**3** pts) $\lim_{x\to 1} \dfrac{x-3}{x^2+2x-4} = \dfrac{\lim_{x\to 1}(x-3)}{\lim_{x\to 1}(x^2+2x-4)}$

True **False**

Why? **Here the denominator limit is $-1 \neq 0$, so the Quotient Law does apply.**

Where points were lost. This was the hardest question on the quiz. Fourteen of you circled True and eleven circled False — and across all 25 papers, **one** person wrote anything at all about the denominator. Both halves miss the same thing. The Quotient Law is the one limit law with a *hypothesis* — Sum, Difference, Product and Constant Multiple have none — and the hypothesis is that the limit of the denominator is not zero. Here it is $1+2-4=-1$, which is not zero, so the law applies and the statement is true. Substituting does not disprove anything; it is how you check the hypothesis. And "because of the limit laws" is not a reason, because it could never come out false: quoting a law with a hypothesis means checking the hypothesis out loud.

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2. (**3** pts) If $f(0.1)$, $f(0.01)$ and $f(0.001)$ all round to 2, then $\lim_{x\to 0} f(x) = 2$.

True **False**

Why? Test $f(x) = \sin\left(\frac{\pi}{x}\right) + 2$: it is exactly 2 at 0.1, 0.01 and 0.001, yet it oscillates between 1 and 3 forever as $x \to 0$. A table samples finitely many points and says nothing about the rest.

Where points were lost. The most common answer was that we are only told about x approaching 0 from the *right*. That is true, but it is not what breaks the statement — fix it by listing $f(-0.1)$, $f(-0.01)$, $f(-0.001)$ as well and the statement is *still* false. Three points are three points, and a limit is a claim about **all** x near 0. The function above passes through exactly 2 at every listed input while swinging between 1 and 3 everywhere in between. No finite table is evidence for a limit — which is also why you were asked to estimate signs rather than plug in values. Thirteen of you circled True. Of the twelve who circled False, eight gave this one-sided reason and **one** gave the reason above — so if this was you, you were with almost the whole room.

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3. (**5** pts) Let

$$g(x) = \begin{cases} 2x+1, & x < 1 \\ 5, & x = 1 \\ 6-3x, & x > 1 \end{cases}$$

(a) $\lim_{x\to 1^-} g(x) =$ 3 (b) $\lim_{x\to 1^+} g(x) =$ 3

(c) $\lim_{x\to 1} g(x) =$ 3 (d) $g(1) =$ 5

(e) Is g continuous at $x = 1$? If not, say which requirement fails.

No. The limit exists and $g(1)$ exists, but they disagree — requirement 3.

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4. (**4** pts) Evaluate the limit. Show the rewriting — substituting first and finding $\frac{0}{0}$ is not an answer, it is the reason a rewrite is needed.

$$\lim_{x\to 0}\frac{\sqrt{x+4}-2}{x}$$

$\frac{1}{4}$ — **multiply by the conjugate** $\frac{\sqrt{x+4}+2}{\sqrt{x+4}+2}$**:**

$$\begin{aligned}\lim_{x\to 0}\frac{\sqrt{x+4}-2}{x} &= \lim_{x\to 0}\frac{\left(\sqrt{x+4}-2\right)\left(\sqrt{x+4}+2\right)}{x\left(\sqrt{x+4}+2\right)}\\ &= \lim_{x\to 0}\frac{(x+4)-4}{x\left(\sqrt{x+4}+2\right)} \;=\; \lim_{x\to 0}\frac{x}{x\left(\sqrt{x+4}+2\right)}\\ &= \lim_{x\to 0}\frac{1}{\sqrt{x+4}+2} \;=\; \frac{1}{\sqrt{4}+2} \;=\; \frac{1}{4}\end{aligned}$$

Where points were lost. Two things, and the first cost a point on its own. **(1) Keep** lim **on every line until you actually take it.** Many of you dropped it after the first step. Without it you have written that a *function* equals a *number*, which is false — and it is not bookkeeping: the cancellation $\frac{x}{x}=1$ is only legal because $x\neq 0$, and the only thing on the page saying $x\neq 0$ is the $\lim_{x\to 0}$. Take the symbol away and you have divided by zero. That is the whole reason 2.3 kept insisting on it. **(2) Algebra.** The conjugate gives $\left(\sqrt{x+4}\right)^2-2^2=(x+4)-4=x$ on top. Common slips were leaving $x+4-2$, squaring only part of the root, or canceling the x against the 4 inside it.

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5. (**3** pts) Let $f(x)=\dfrac{x+1}{x-3}$. **No calculator.** Decide each limit by naming the sign and size of the numerator and of the denominator.

(a) $\lim_{x\to 3^-} f(x) =$ $-\infty$ (b) $\lim_{x\to 3^+} f(x) =$ $+\infty$

(c) Write the equation of the vertical asymptote.

$x=3$

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Just left of 3 the numerator is near 4 (positive) and $x-3$ is small and **negative**, so the quotient is large and negative. Just right of 3 the denominator is small and positive.

6. (**2** pts) Find the value of the constant c that makes

$$h(x) \;=\; \begin{cases} x^2-1, & x<3\\ cx+2, & x\geq 3\end{cases}$$

continuous at $x=3$.

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Both pieces are polynomials, so each is continuous on its own side and the only place continuity can fail is the junction. The pieces therefore just have to meet at $x=3$:

$$3^2-1 \;=\; 3c+2 \qquad\Longrightarrow\qquad 8=3c+2 \qquad\Longrightarrow\qquad \mathbf{c=2}$$